

FMUSER FUQ-4655BC Encoder Modulator 24 CVBS & 512 IP Inputs 16 RF Outputs MPEG-2 Encoding DVB-C-DVB-T-ATSC-ISDB-T Multiplexing Scrambling CATV Headend Hotel TV System



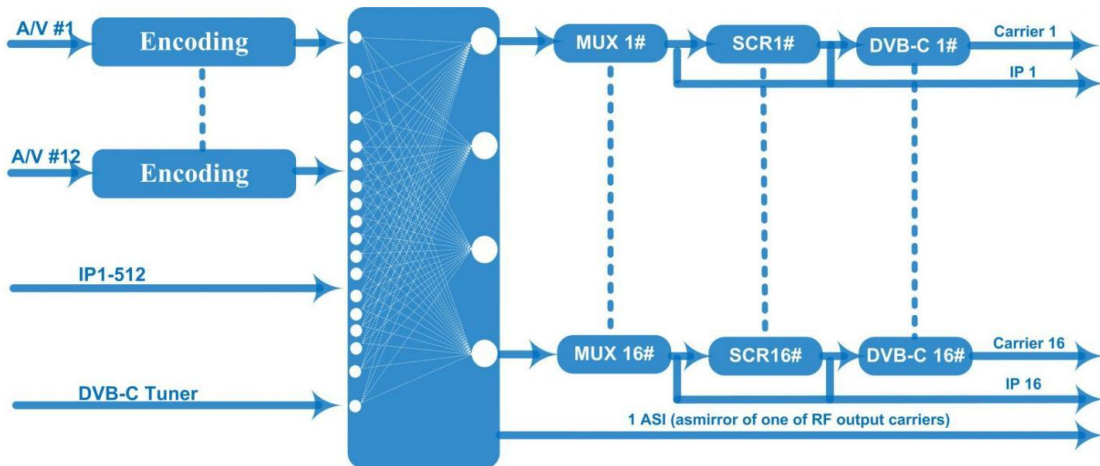
Product Overview

The FMUSER FUQ-4655BC is a professional SD encoder modulator integrating encoding, multiplexing, scrambling, and modulation, with 24 CVBS inputs and IP streaming via the DATA1 port. It processes signals using MPEG-2 video encoding and MPEG-1 Layer 2 or DD AC3 audio encoding, outputting DVB-C, DVB-T/ATSC, or ISDB-T signals. Featuring web-based management for control and updates, it is ideal for small CATV headends and broadcast systems, enabling efficient signal distribution.

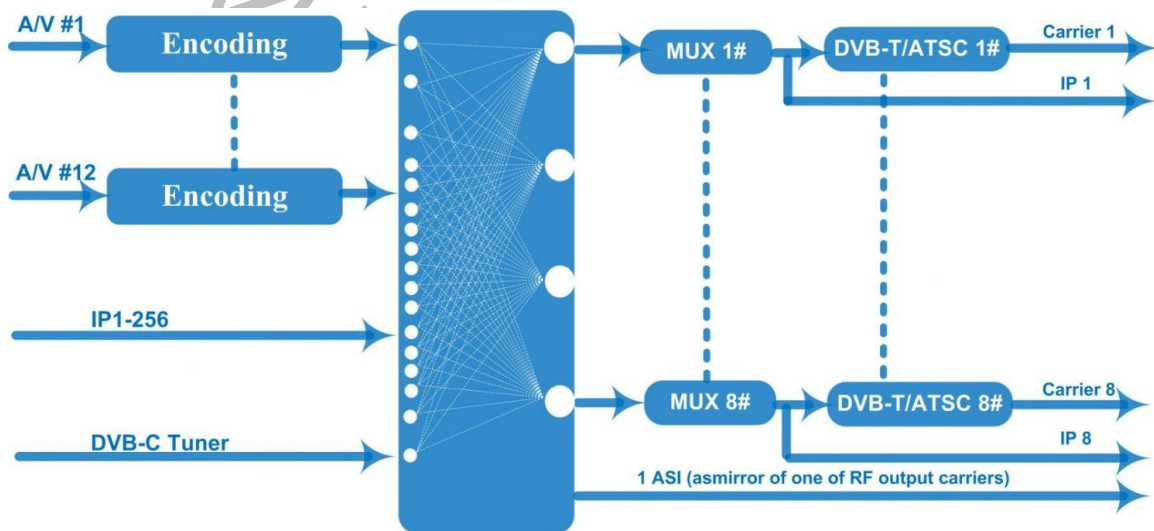
Key feature

Comprehensive multiplexing with PID remapping, PCR adjusting, and PSI/SI editing.

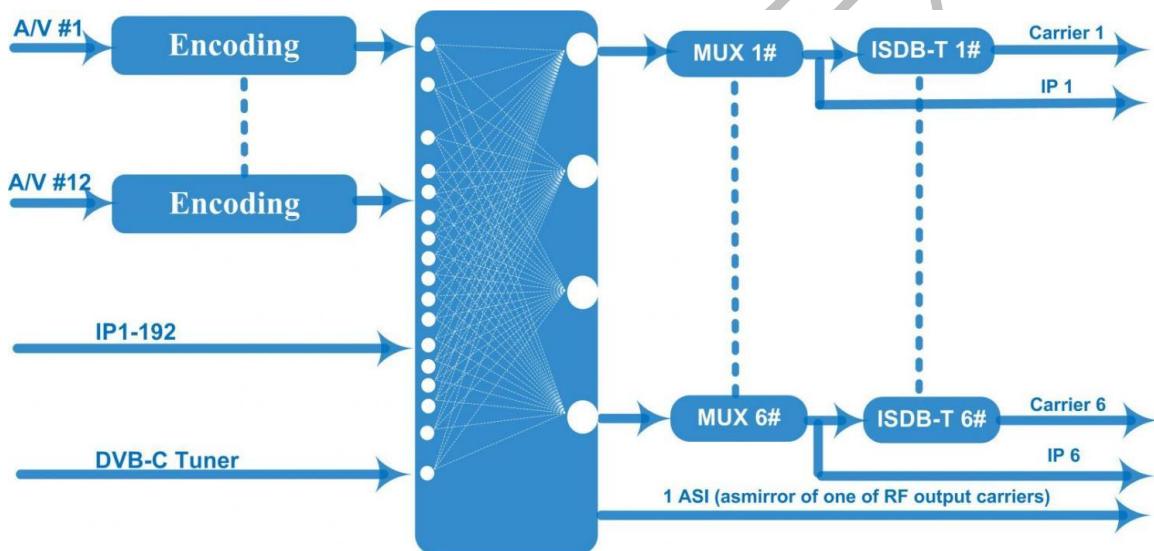
Principle Chart



DVB-C Principle Chart



DVB-T Principle Chart



ISDB-T Principle Chart

Technical specification

Input	24 CVBS inputs 1 DVB-C(ATSC optional) Tuner for re-mux, F type interface 512 IP input over UDP and RTP, DATA1, RJ45—DVB-C RF out 256 IP input over UDP and RTP, DATA1, RJ45—DVB-T/ATSC RF out 192 IP input over UDP and RTP, DATA1, RJ45—ISDB RF out	24 CVBS inputs 1 DVB-C(ATSC optional) Tuner for re-mux, F type interface 512 IP input over UDP and RTP, DATA1, RJ45—DVB-C RF out 256 IP input over UDP and RTP, DATA1, RJ45—DVB-T/ATSC RF out 192 IP input over UDP and RTP, DATA1, RJ45—ISDB RF out	24 CVBS inputs 1 DVB-C(ATSC optional) Tuner for re-mux, F type interface 512 IP input over UDP and RTP, DATA1, RJ45—DVB-C RF out 256 IP input over UDP and RTP, DATA1, RJ45—DVB-T/ATSC RF out 192 IP input over UDP and RTP, DATA1, RJ45—ISDB RF out	24 CVBS inputs 1 DVB-C(ATSC optional) Tuner for re-mux, F type interface 512 IP input over UDP and RTP, DATA1, RJ45—DVB-C RF out 256 IP input over UDP and RTP, DATA1, RJ45—DVB-T/ATSC RF out 192 IP input over UDP and RTP, DATA1, RJ45—ISDB RF out	24 CVBS inputs 1 DVB-C(ATSC optional) Tuner for re-mux, F type interface 512 IP input over UDP and RTP, DATA1, RJ45—DVB-C RF out 256 IP input over UDP and RTP, DATA1, RJ45—DVB-T/ATSC RF out 192 IP input over UDP and RTP, DATA1, RJ45—ISDB RF out
Video	Resolution	720×480_60i, 544×480_60i, 352×480_60i, 352×240_60i, 320×240_60i, 176×240_60i, 176×120_60i	720×480_60i, 544×480_60i, 352×480_60i, 352×240_60i, 320×240_60i, 176×240_60i, 176×120_60i	720×480_60i, 544×480_60i, 352×480_60i, 352×240_60i, 320×240_60i, 176×240_60i, 176×120_60i	720×480_60i, 544×480_60i, 352×480_60i, 352×240_60i, 320×240_60i, 176×240_60i, 176×120_60i
Video	Resolution	720×576_50i, 704×576_50i, 640×576_50i, 352×288_50i, 320×288_50i, 176×288_50i, 176×144_50i	720×576_50i, 704×576_50i, 640×576_50i, 352×288_50i, 320×288_50i, 176×288_50i, 176×144_50i	720×576_50i, 704×576_50i, 640×576_50i, 352×288_50i, 320×288_50i, 176×288_50i, 176×144_50i	720×576_50i, 704×576_50i, 640×576_50i, 352×288_50i, 320×288_50i, 176×288_50i, 176×144_50i
Video	Encoding	MPEG-2	MPEG-2	MPEG-2	MPEG-2
Video	Bit-rate	0.5Mbps~8M bps per channel	0.5Mbps~8M bps per channel	0.5Mbps~8M bps per channel	0.5Mbps~8M bps per channel
Video	Rate Control	CBR/VBR	CBR/VBR	CBR/VBR	CBR/VBR
Video	GOP Structure	IP, IBP, IBBP, IBBBBP	IP, IBP, IBBP, IBBBBP	IP, IBP, IBBP, IBBBBP	IP, IBP, IBBP, IBBBBP

Video	Advanced Pretreatment	De-interlacing , noise reduction	De-interlacing , noise reduction	De-interlacing , noise reduction	De-interlacing , noise reduction
Audio	Encoding	MPEG-1 Layer 2, DD AC3 (2.0)	MPEG-1 Layer 2, DD AC3 (2.0)	MPEG-1 Layer 2, DD AC3 (2.0)	MPEG-1 Layer 2, DD AC3 (2.0)
Audio	Sampling rate	48KHz	48KHz	48KHz	48KHz
Audio	Resolution	24-bit	24-bit	24-bit	24-bit
Audio	Bit-rate	64Kbps,128Kbps,192kbps,256kbps,320kbps,384kbps each channel	64Kbps,128Kbps,192kbps,256kbps,320kbps,384kbps each channel	64Kbps,128Kbps,192kbps,256kbps,320kbps,384kbps each channel	64Kbps,128Kbps,192kbps,256kbps,320kbps,384kbps each channel
Multiplexing	Maximum PID Remapping	180 input per channel	180 input per channel	180 input per channel	180 input per channel
Multiplexing	Function	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)
Multiplexing	Function	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting
Multiplexing	Function	Generate PSI/ SI table automatically	Generate PSI/ SI table automatically	Generate PSI/ SI table automatically	Generate PSI/ SI table automatically
Scrambling	Maximum simulcrypt CA	4	4	4	4
Scrambling	Standard	ETR289, ETSI 101 197, ETSI 103 197	ETR289, ETSI 101 197, ETSI 103 197	ETR289, ETSI 101 197, ETSI 103 197	ETR289, ETSI 101 197, ETSI 103 197
Scrambling	Connection	Local/remote connection	Local/remote connection	Local/remote connection	Local/remote connection
Modulation	DVB-C	QAM Channel	QAM Channel	16 non-adjacent carriers output (maximum bandwidth	16 non-adjacent carriers output (maximum bandwidth

Modulation	DVB-C	Standard	Standard	EN300 429/ITU-T J.83A/B	EN300 429/ITU-T J.83A/B
Modulation	DVB-C	MER	MER	≥40db	≥40db
Modulation	DVB-C	RF frequency	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulation	DVB-C	RF output level	RF output level	-20~+10dbm (87~107 dbμV), 0.1db step	-20~+10dbm (87~107 dbμV), 0.1db step
Modulation	DVB-C	Symbol Rate	Symbol Rate	5.0Msps~7.0 Msps, 1ksps stepping	5.0Msps~7.0 Msps, 1ksps stepping
Modulation	DVB-C			J.83A	J.83B
Modulation	DVB-C	Constellation	Constellation	16/32/64/128/ 256QAM	64/256 QAM
Modulation	DVB-C	Bandwidth	Bandwidth	8M	6M
Modulation	DVB-T	Standard	Standard	EN300744	EN300744
Modulation	DVB-T	FFT mode	FFT mode	2K,4K, 8K	2K,4K, 8K
Modulation	DVB-T	Bandwidth	Bandwidth	6M, 7M, 8M	6M, 7M, 8M
Modulation	DVB-T	Constellation	Constellation	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
Modulation	DVB-T	Guard Interval	Guard Interval	1/4, 1/8, 1/16, 1/32	1/4, 1/8, 1/16, 1/32
Modulation	DVB-T	FEC	FEC	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
Modulation	DVB-T	MER	MER	≥42 dB	≥42 dB
Modulation	DVB-T	RF frequency	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulation	DVB-T	RF out	RF out	8 non-adjacent carriers DVB-T out (maximum bandwidth 192MHz)	8 non-adjacent carriers DVB-T out (maximum bandwidth 192MHz)

Modulation	DVB-T	RF output level	RF output level	-20~ +10 dBm (77~97 dbμV), 0.1db step	-20~ +10 dBm (77~97 dbμV), 0.1db step
Modulation	ATSC	Standard	Standard	ATSC A/53	ATSC A/53
Modulation	ATSC	Bandwidth	Bandwidth	6M	6M
Modulation	ATSC	Constellation	Constellation	8VSB	8VSB
Modulation	ATSC	FEC	FEC	RS(208 188)+Trellis	RS(208 188)+Trellis
Modulation	ATSC	MER	MER	≥40dB	≥40dB
Modulation	ATSC	ACL	ACL	-55 dBc	-55 dBc
Modulation	ATSC	RF frequency	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulation	ATSC	RF out	RF out	8 non-adjacent carriers output (maximum bandwidth 192MHz)	8 non-adjacent carriers output (maximum bandwidth 192MHz)
Modulation	ATSC	RF output level	RF output level	-20~+10dbm (for all carriers), 0.5db stepping	-20~+10dbm (for all carriers), 0.5db stepping
Modulation	ISDB-T	Standard	Standard	ARIB STD-B31	ARIB STD-B31
Modulation	ISDB-T	Bandwidth	Bandwidth	6M	6M
Modulation	ISDB-T	Constellation	Constellation	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
Modulation	ISDB-T	Guard Interval	Guard Interval	1/32, 1/16, 1/8, 1/4	1/32, 1/16, 1/8, 1/4
Modulation	ISDB-T	Transmission Mode	Transmission Mode	2K, 4K, 8K	2K, 4K, 8K
Modulation	ISDB-T	Code rate	Code rate	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
Modulation	ISDB-T	MER	MER	≥40dB	≥40dB

Modulation	ISDB-T	RF frequency	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulation	ISDB-T	RF out	RF out	6 non-adjacent carriers output (maximum bandwidth 192MHz)	6 non-adjacent carriers output (maximum bandwidth 192MHz)
Modulation	ISDB-T	RF output level	RF output level	-20dBm~+10 dBm, 0.1dB stepping	-20dBm~+10 dBm, 0.1dB stepping
Stream output	1 ASI output as mirror of one of RF output carrier s-Optional 16 MPTS output over UDP and RTP/RTSP as mirror of 16 DVB-C carriers, 8 MPTS output over UDP and RTP/RTSP as mirror of 8 DVB-T/ATSC carriers(Optional) 6 MPTS output over UDP and RTP/RTSP as mirror of 6 ISDB-T carriers(Optional) 1*1000M Base-T Ethernet interface	1 ASI output as mirror of one of RF output carrier s-Optional 16 MPTS output over UDP and RTP/RTSP as mirror of 16 DVB-C carriers, 8 MPTS output over UDP and RTP/RTSP as mirror of 8 DVB-T/ATSC carriers(Optional) 6 MPTS output over UDP and RTP/RTSP as mirror of 6 ISDB-T carriers(Optional) 1*1000M Base-T Ethernet interface	1 ASI output as mirror of one of RF output carrier s-Optional 16 MPTS output over UDP and RTP/RTSP as mirror of 16 DVB-C carriers, 8 MPTS output over UDP and RTP/RTSP as mirror of 8 DVB-T/ATSC carriers(Optional) 6 MPTS output over UDP and RTP/RTSP as mirror of 6 ISDB-T carriers(Optional) 1*1000M Base-T Ethernet interface	1 ASI output as mirror of one of RF output carrier s-Optional 16 MPTS output over UDP and RTP/RTSP as mirror of 16 DVB-C carriers, 8 MPTS output over UDP and RTP/RTSP as mirror of 8 DVB-T/ATSC carriers(Optional) 6 MPTS output over UDP and RTP/RTSP as mirror of 6 ISDB-T carriers(Optional) 1*1000M Base-T Ethernet interface	1 ASI output as mirror of one of RF output carrier s-Optional 16 MPTS output over UDP and RTP/RTSP as mirror of 16 DVB-C carriers, 8 MPTS output over UDP and RTP/RTSP as mirror of 8 DVB-T/ATSC carriers(Optional) 6 MPTS output over UDP and RTP/RTSP as mirror of 6 ISDB-T carriers(Optional) 1*1000M Base-T Ethernet interface
System function	Network management (WEB)	Network management (WEB)	Network management (WEB)	Network management (WEB)	Network management (WEB)

System function	Chinese and English language	Chinese and English language	Chinese and English language	Chinese and English language	Chinese and English language
System function	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade
Miscellaneous	Dimension (W×L×H)	Dimension (W×L×H)	482mm×440mm×44mm	482mm×440mm×44mm	482mm×440mm×44mm
Miscellaneous	Environment	Environment	0~45■(work) ■-20~80■S storage■	0~45■(work) ■-20~80■S storage■	0~45■(work) ■-20~80■S storage■
Miscellaneous	Power requirements	Power requirements	AC 110V±10%, 50/60Hz, AC 220 ± 10%, 50/60Hz	AC 110V±10%, 50/60Hz, AC 220 ± 10%, 50/60Hz	AC 110V±10%, 50/60Hz, AC 220 ± 10%, 50/60Hz

Ordering Information

	IP input	IP output	RF carriers	Scrambling
DVB-C	512	16	16	√
DVB-T	256	8	8	×
ATSC	256	8	8	×
ISDB-T	192	6	6	×

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

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